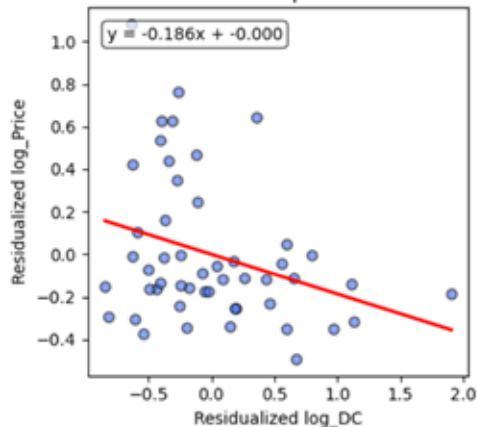


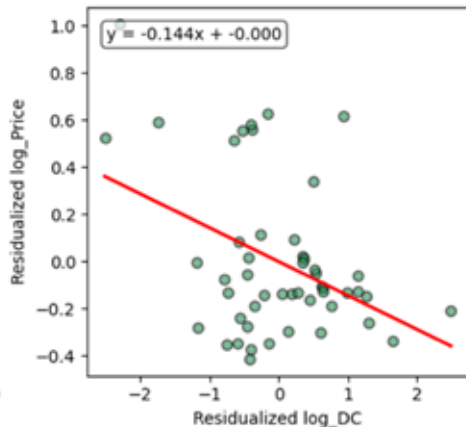
Does data center capacity per state predict state-level electricity prices while controlling for population, urbanization, and natural gas consumption?

Matthew Ketas
Ivan Zheng

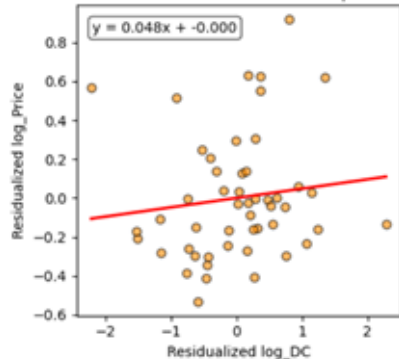
Control: Population



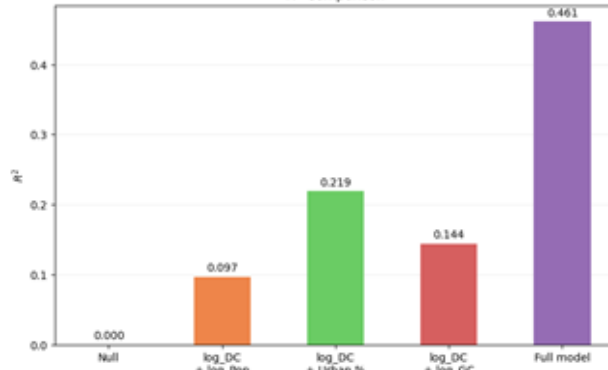
Control: Urban Percent



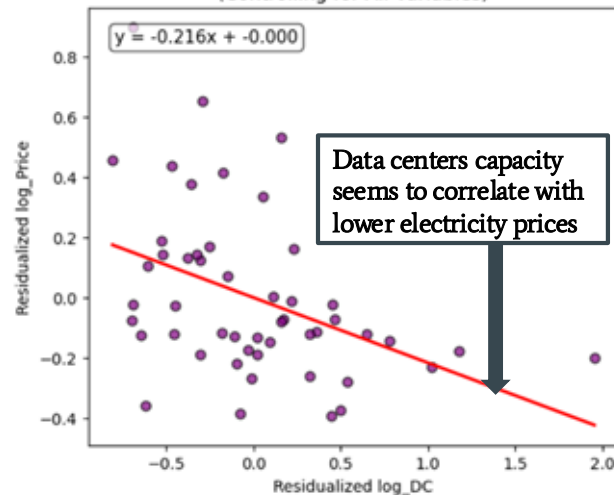
Control: Natural Gas Consumption



R² Comparison



Effect of Data Center Capacity on Electricity Prices
(Controlling for All Variables)



$$\text{Electricity Price} = \beta_0 + \beta_1(\text{DataCenterCapacity}) + \beta_2(\text{Population}) + \beta_3(\% \text{Urban}) + \beta_4(\text{NaturalGasConsumption}) + \varepsilon$$

--- Log Pop only Model ---

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.6202	1.149	0.540	0.592	-1.691	2.931
log_DC	-0.1865	0.083	-2.242	0.030	-0.354	-0.019
log_Pop	0.1740	0.093	1.874	0.067	-0.013	0.361

Strong and
Negative

--- Log DC only Model ---

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.2137	0.222	9.959	0.000	1.767	2.661
log_DC	-0.1439	0.047	-3.041	0.004	-0.239	-0.049
Urban_Percent	0.0123	0.004	3.376	0.001	0.005	0.020

--- Log Total Price Model ---

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.3985	0.910	0.438	0.664	-1.434	2.231
log_DC	-0.2160	0.069	-3.110	0.003	-0.356	-0.076
log_Pop	0.3088	0.083	3.722	0.001	0.142	0.476
Urban_Percent	0.0094	0.003	2.954	0.005	0.003	0.016
log_GC	-0.1875	0.047	-3.971	0.000	-0.283	-0.092

--- Log GC only Model ---

	coef	std err	t	P> t	[0.025	0.975]
Intercept	4.0022	0.525	7.624	0.000	2.946	5.058
log_DC	0.0479	0.057	0.848	0.401	-0.066	0.162
log_GC	-0.1279	0.051	-2.510	0.016	-0.230	-0.025

Weak and
Positive

Takeaways:

- Data centers are most likely built where energy is cheap.
- Energy markets are very diverse or omitted variable bias
- Short term spikes aren't recorded in the data